FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Child Survival & Key Interventions in Uganda

Lecture | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Mawanda

Uganda's under-five mortality has fallen to about 52 per 1,000 live births (UDHS 2022), but around 40% of these deaths are now neonatal, and malaria, pneumonia, diarrhoea and undernutrition still kill most older children. A short list of cheap, proven interventions (skilled birth care, breastfeeding, vaccines, nets, ORS and zinc, amoxicillin, ACTs, vitamin A, KMC) delivered through IMCI, iCCM and ETAT could prevent most of them.

Must know before the ward round

1. Definitions: neonatal mortality rate (NMR) = deaths 0–28 days per 1,000 live births; infant mortality rate (IMR) = deaths under 1 year per 1,000 live births; under-five mortality rate (U5MR) = deaths under 5 years per 1,000 live births.
2. Uganda UDHS 2022: NMR about 22, IMR about 36, U5MR about 52 per 1,000 live births (UDHS 2016: 27, 43, 64). SDG 3.2 targets for 2030: NMR ≤12 and U5MR ≤25.
3. Globally (WHO, 2024 data): 4.9 million under-five deaths; 47% are neonatal; U5MR 37.4 per 1,000; sub-Saharan Africa 71.6 per 1,000 and 58% of all under-five deaths.
4. Leading causes of under-five death: preterm birth complications, intrapartum events (birth asphyxia), pneumonia, diarrhoea and malaria; in Uganda malaria remains the leading cause of illness and death in young children. Undernutrition underlies about 45% of child deaths.
5. Leading neonatal killers: prematurity, intrapartum asphyxia and sepsis (with pneumonia), about 8 in 10 newborn deaths. Most newborn deaths occur in the first week, many on day 1.
6. IMCI (WHO/UNICEF, 1995) has three components: improve health-worker case management skills, strengthen health systems, and improve family and community practices. It follows the sequence: assess → classify (pink/yellow/green) → identify treatment → treat → counsel → follow up.
7. General danger signs (2 months–5 years): unable to drink or breastfeed, vomits everything, convulsions (now or in this illness), lethargic or unconscious. Any one = urgent referral after pre-referral treatment.
8. iCCM (Uganda since 2010): VHTs treat children 2–59 months for malaria (RDT + ACT), pneumonia (amoxicillin DT 250 mg: 2–11 months 1 tab, 1–5 years 2 tabs, 12-hourly for 5 days) and diarrhoea (ORS + zinc 10 days), give pre-referral rectal artesunate and refer danger signs.
9. Zinc for diarrhoea (UCG 2023): 2–6 months 10 mg (½ tab) and 6 months–5 years 20 mg (1 tab) once daily for 10 days, with low-osmolarity ORS (WHO 2024 suggests 5 mg daily).
10. Lancet Child Survival Series (2003): about two-thirds (63%) of child deaths could be prevented with 23 existing low-cost interventions at full coverage; breastfeeding, ORS, ITNs, antibiotics for pneumonia and sepsis, and vaccines head the list.
11. Newborn high-impact package: skilled birth attendance, Helping Babies Breathe (the Golden Minute), immediate drying and skin-to-skin, delayed cord clamping, breastfeeding within 1 hour, chlorhexidine cord care, vitamin K, eye prophylaxis, Kangaroo Mother Care (iKMC for all <2 kg, WHO 2022), and early treatment of sepsis.
12. GOBI-FFF (UNICEF): Growth monitoring, Oral rehydration, Breastfeeding, Immunisation; Female education, Family spacing, Food supplementation.

1 Definitions and Key Concepts
2 Burden and Uganda Context
3 Principles and Framework
4 Application in Practice

5 Clinical Acumen
6 Common Problems and Their Management
7 Guidelines, Policy and Law in Uganda
8 Key Points
9 References

1. Definitions and Key Concepts

Child survival is about preventing the deaths of children under 5. Ugandan children are far more likely to survive now than in 1990, but about **1 in 19** still dies before the fifth birthday, and **progress has slowed**, especially for **newborns**.

The causes are not mysterious. Most deaths come from a **few conditions**: preterm birth, birth asphyxia, neonatal sepsis, **malaria**, **pneumonia** and **diarrhoea**, made deadlier by **malnutrition** and **HIV**. Each has a **cheap, proven intervention**. Children die because those interventions **don't reach them**, or reach them **too late**: the family delays seeking care, the child can't get to a facility, or the facility can't give quality care.

Uganda's main strategies are:

- the **continuum of care** (adolescence → pregnancy → birth → newborn → childhood);
- **IMCI** at facilities, **iCCM** by Village Health Teams (VHTs), and **ETAT** in hospitals;
- **prevention**: immunisation (now including **malaria vaccine**), ITNs, breastfeeding, nutrition, WASH;
- **accountability**: maternal and perinatal death surveillance and response (**MPDSR**).

> **CLINICAL PEARL**
> The one thing never to forget: **a child who dies of pneumonia, diarrhoea or malaria almost always had a treatable illness for which care arrived too late.** On every ward round, ask "When did it start? When did they seek care? What happened on the way?" (the **three delays**).

Mortality and nutrition indicators

Indicator	Definition	Uganda (UDHS 2022, approx.)
Neonatal mortality rate (NMR)	Deaths in the first **28 days** per 1,000 live births	**22**
Post-neonatal mortality	Deaths from 1 to 11 months per 1,000 live births	About 14 (IMR − NMR)
Infant mortality rate (IMR)	Deaths before the **first birthday** per 1,000 live births	**36**
Child mortality rate	Deaths between 1 and 4 years per 1,000 children surviving to age 1	About 17
Under-five mortality rate (U5MR)	Probability of dying before age **5** per 1,000 live births	**52**
Perinatal mortality	Stillbirths (≥28 weeks) + early neonatal deaths (first 7 days) per 1,000 total births	Around 30–40 Verify *UDHS 2022 perinatal mortality rate*
Stillbirth rate	Babies born with no signs of life at ≥28 weeks per 1,000 total births	Not given in the source notes
Stunting	Height-for-age < −2 Z (chronic undernutrition)	About **24%** of under-5s (UDHS 2022; 29% in 2016)
Wasting	Weight-for-height < −2 Z or MUAC <12.5 cm (acute)	Low nationally (about 3–4%) but much higher in **Karamoja**

Indicator	Definition	Uganda (UDHS 2022, approx.)
Zero-dose child	No DTP1	36,452 in 2025 (MoH)

SDG 3.2 (by 2030): end preventable deaths of newborns and under-fives; every country to reduce **NMR to ≤12** and **U5MR to ≤25** per 1,000 live births.

> NOTE
> UDHS figures are averages over the 5 years before the survey; check the UDHS 2022 final report for exact values and confidence intervals. Different sources (UN IGME modelled estimates, UDHS) give slightly different numbers. In an exam, quote one source and its year. Remember that U5MR is a **probability per 1,000 live births**, not deaths per 1,000 children under 5.

Key concepts

Concept	Meaning
Continuum of care	Linked care across life stages (adolescence, pregnancy, birth, newborn, childhood) and across levels (home, community, facility)
The three delays	(1) Deciding to seek care, (2) reaching care, (3) receiving adequate care once there
IMCI / IMNCI	Integrated Management of (Newborn and) Childhood Illness: assess, classify by colour, treat, counsel, follow up
iCCM	Integrated community case management: VHTs treat malaria, pneumonia and diarrhoea in children 2–59 months
ETAT	Emergency Triage Assessment and Treatment in hospitals
Missed opportunity	Any contact where a due vaccine, vitamin A, growth check, HIV test or counselling was not given

2. Burden and Uganda Context

Global picture (WHO fact sheet, data for 2024)

- **4.9 million** children died before age 5 in 2024: one every 6 seconds.
- **2.3 million (47%)** were **neonatal** deaths.
- Global **U5MR 37.4** per 1,000 (less than half the 2000 level), but progress has **slowed since 2015** and the number of deaths has stagnated since 2022.
- **Sub-Saharan Africa: U5MR 71.6** per 1,000, nearly 14 times higher than Europe/North America, and **58%** of all under-five deaths.
- Leading causes: **preterm complications, intrapartum events, pneumonia, diarrhoea, malaria**.
- Leading **neonatal** killers: prematurity, intrapartum asphyxia and sepsis (with pneumonia), about 8 in 10 newborn deaths; most occur in the **first week**, many on **day 1**.

Uganda

- **Trend:** U5MR fell from about 90 (UDHS 2011) to **64 (2016)** to **52 (2022)**. Neonatal mortality has fallen much more slowly (27 → **22**), so the **newborn share** of under-five deaths keeps rising (now about 40%).
- **Causes:**
  - Newborns: **prematurity**, **birth asphyxia**, **sepsis/pneumonia**, congenital anomalies.

    - 1–59 months: **malaria** (MoH: "the leading cause of illness and death of young children in Uganda"), **pneumonia**, **diarrhoea**, plus **severe anaemia**, **HIV**, **measles** (outbreaks in 66 districts in 2025), **injuries** (burns, drowning, road traffic), and **sickle cell disease**.
    - **Undernutrition** (stunting, wasting, micronutrient deficiency) is an underlying cause in nearly half of deaths (about **45%**).

- **Inequity:** mortality is higher in **rural** areas, in the **poorest** quintile, among children of mothers with **no education**, in **Karamoja**, West Nile and refugee-hosting districts, among children born **<24 months** after a sibling, and to **adolescent mothers**.
- **Where children die:** many at **home** or **in transit**; in hospitals, most deaths happen within **24–48 hours of admission**, a sign of late arrival and of triage delays.

> **UGANDA CONTEXT**
> The health-system ladder: **VHT (iCCM)** → **HC II** (parish; outpatient) → **HC III** (sub-county; maternity, lab) → **HC IV** (county; theatre, blood) → **general hospital** → **regional referral hospital** → **national referral hospital** (Mulago, and specialised units). Mission hospitals such as **Mengo** and **Nsambya** (Uganda Protestant/Catholic Medical Bureaus) provide a large share of care. Common gaps: stock-outs (amoxicillin DT, zinc, ACTs), **oxygen** and pulse oximeters, blood, transport for referral, and staffing at night.

3. Principles and Framework

The causal chain

Distal (basic): poverty, maternal education, gender, conflict, health financing

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Intermediate (underlying): food insecurity, poor WASH, crowding, no ITN, short birth spacing, low vaccine coverage, care-seeking practices

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Proximate: low birth weight, malnutrition, HIV, exposure to pathogens

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Immediate: pneumonia, diarrhoea, malaria, neonatal sepsis, asphyxia, measles

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Three delays: deciding to seek care, reaching care, receiving quality care

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Death or survival with disability

(Adapted from the Mosley–Chen framework and the UNICEF conceptual framework of malnutrition.) Every level is a point of intervention: girls' education and family planning at the top, nets, WASH and vaccines in the middle, case management and the three delays at the bottom.

Why malnutrition multiplies the risk

Undernutrition weakens the gut barrier, cell-mediated immunity and the response to infection, and infection in turn worsens malnutrition (the **infection–malnutrition cycle**). A wasted child with pneumonia or diarrhoea is many times more likely to die than a well-nourished child with the same illness.

Proven, cheap interventions

The **Lancet Child Survival Series (2003)** estimated that about two-thirds (**63%**) of child deaths could be prevented with **23** existing low-cost interventions at full coverage; breastfeeding, ORS, ITNs, antibiotics for pneumonia and sepsis, and vaccines head the list. UNICEF's **GOBI-FFF** is the older shorthand: **G**rowth

monitoring, **O**ral rehydration, **B**reastfeeding, **I**mmunisation; **F**emale education, **F**amily spacing, **F**ood supplementation.

IMCI: the core clinical framework

IMCI (WHO/UNICEF, 1995) has **three components**: improve health-worker case management skills, strengthen health systems, and improve family and community practices. The clinical sequence is **assess → classify (pink/yellow/green) → identify treatment → treat → counsel → follow up**.

Colour	Meaning	Action
Pink	Severe classification	Pre-referral treatment and urgent referral (or admission)
Yellow	Specific treatment needed	Treat at the facility or home, counsel, follow up
Green	Home care	Counsel on home care and when to return

Key frameworks and programmes

Framework	Year	Core idea
Primary Health Care (Alma-Ata)	1978	Health for all through community-based essential care
GOBI-FFF (UNICEF)	1980s	Growth monitoring, ORS, Breastfeeding, Immunisation; Female education, Family spacing, Food supplementation
EPI (UNEPI in Uganda)	1974 (global); 1983 (UNEPI)	Routine immunisation
IMCI (WHO/UNICEF)	1995 (Uganda late 1990s); now IMNCI including newborns	Integrated assessment and treatment of the sick child
Lancet Child Survival Series	2003	23 interventions could prevent 63% of deaths
ETAT (WHO)	2005; updated 2016	Triage and emergency treatment in hospitals
iCCM (Uganda)	2010	VHTs treat malaria, pneumonia, diarrhoea in the community
A Promise Renewed / RMNCAH Sharpened Plan	2013; updated	Uganda's investment plan for RMNCAH
GAPPD (WHO/UNICEF)	2013	Protect, Prevent, Treat pneumonia and diarrhoea
Every Newborn Action Plan (ENAP)	2014	Coverage targets for newborn care and stillbirth prevention
Nurturing Care Framework	2018	Survive, thrive and transform: health, nutrition, responsive care, early learning, safety
WHO guideline on pneumonia and diarrhoea in children up to 10 years	**2024**	Current WHO case management standards

4. Application in Practice

Child survival is public health, but it is practised **one child at a time**. The clinical skill is **IMCI-style assessment**: picking out the child who will die without urgent action.

Step 0: First 5 minutes (ETAT triage at the door)

1. **E**mergency signs (ABCD: Airway/Breathing, Circulation, Coma/Convulsions, Dehydration) → treat immediately in the emergency area.
2. **P**riority signs ("3 TPR-MOB": **T**iny baby <2 months, **T**emperature very high, **T**rauma; **P**allor (severe), **P**oisoning, **P**ain (severe); **R**estless/irritable/lethargic, **R**espiratory distress, **R**eferral (urgent); **M**alnutrition (visible severe wasting), **O**edema of both feet, **B**urns (major)) → assess and treat next.
3. **Q**ueue (non-urgent) → routine IMCI.
4. Check **RBS** and **SpO_2** in every sick child; give oxygen if SpO_2 <90%; treat hypoglycaemia (glucose <2.5 mmol/L, or <3 mmol/L in malnourished) with **10% dextrose 5 mL/kg IV** (or sugar water by NGT).

Step 1: Focused history (for every sick child under 5)

Ask	Why it matters
Age, and **how long ill**	Infants and long illnesses carry higher risk
General danger signs: drinking/breastfeeding? vomiting everything? convulsions?	Any one = urgent referral (pink)
Cough or difficult breathing; diarrhoea (how long, blood?); fever (how long, malaria test?); ear problem	The four IMCI main symptoms; classification and treatment
Where did they go first? (drug shop, herbalist, VHT, clinic) and what was given	First delay, prior treatment, partial antibiotics
Feeding (breastfeeding, complementary foods, appetite)	Malnutrition risk
Immunisation (card) and **vitamin A**, deworming	Missed opportunities to close today
HIV status of mother and child	HIV changes thresholds and treatment
ITN use last night	Malaria prevention
Birth history for infants (preterm, LBW, asphyxia)	Newborn risk
Social: who cares for the child, distance to facility, money for transport	Second delay; can this child be treated at home and come back?

Step 2: Focused examination (Hutchison order, IMCI content)

1. **Introduce yourself**, obtain the carer's consent, wash your hands, and examine the child on the mother's lap where possible; **observe first** while the child is calm.
2. **General inspection from a distance** ("look, listen, feel"): danger signs (lethargic, unconscious, convulsing), well or unwell, distress, colour, hydration, visible severe wasting.
3. **Vital signs**: count **RR for a full minute** while the child is quiet (fast: ≥60 if <2 months; ≥50 at 2–11 months; ≥40 at 12–59 months), **SpO_2** (hypoxaemia <90%), temperature, pulse.
4. **Growth and nutrition**: weight, height/length, weight-for-height/length **Z-score**, **MUAC** (<11.5 cm severe, 11.5–12.4 cm moderate), and bilateral **pitting oedema** of the feet. Plot on the growth chart.
5. **Hands**: **palmar pallor** (severe or some), capillary refill, temperature of the hands (shock).
6. **Face and eyes**: sunken eyes (dehydration), conjunctival pallor, measles signs (red eyes, coryza).
7. **Neck**: stiff neck (meningitis).
8. **Skin**: **skin pinch** (dehydration), measles rash, burns, signs of abuse.
9. **Chest**: chest indrawing, stridor in a calm child, wheeze.
10. **Abdomen**: as indicated (spleen and liver in fever and pallor).
11. **Ears last**: discharge, tender swelling behind the ear (mastoiditis).
12. Thank the carer, wash your hands, classify by colour and **summarise**.

Step 3: Bedside tests

Test	Use	Why it matters
Malaria RDT (or smear)	Every febrile child in Uganda (test, then treat)	Avoids treating every fever as malaria
SpO_2	Hypoxaemia → oxygen and referral	The single strongest predictor of death in pneumonia
RBS	Hypoglycaemia in the very sick, malnourished, convulsing	Common, deadly and easily treated
Hb (HemoCue)	Severe anaemia	Transfusion decision
MUAC tape, weight, height, oedema	Acute malnutrition (6–59 months)	Changes fluid, antibiotic and feeding management
Respiratory timer (ARI timer)	Counting RR accurately (VHTs and HCs)	Fast breathing defines pneumonia
HIV rapid test	Unknown status in a sick or malnourished child	ART, cotrimoxazole, different thresholds
Immunisation card	Missed doses	Catch up before discharge

Step 4: Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management	Availability
Blood smear for malaria	Severe malaria suspected; RDT negative but clinical suspicion	Parasitaemia, species	Confirms; follow parasite clearance	HC III and above
FBC	Pallor, sepsis, severe illness	Severe anaemia, leucocytosis	Transfusion; antibiotics	HC IV and above
Blood culture	Sepsis, severe pneumonia	Organism and sensitivities	Targeted antibiotics	Regional/national referral; few HC IVs
Lumbar puncture	Convulsions with fever, stiff neck, bulging fontanelle	Meningitis	Antibiotics for 10–14 days	Hospitals
Stool microscopy	Persistent diarrhoea, dysentery	Parasites (e.g. *Giardia*, amoeba)	Specific treatment	HC III and above
Electrolytes, creatinine	Severe dehydration, SAM, AKI	Hypo-/hypernatraemia, hypokalaemia	Fluid choice	Hospitals
Chest X-ray	Severe pneumonia not responding, TB suspected	Consolidation, effusion, hilar nodes	Change antibiotics, drain, TB treatment	General hospital and above
GeneXpert	TB symptoms or contact, SAM not responding	MTB detected	TB treatment	HC IV and above

Step 5: Putting it together

Model one-liner: "A 14-month-old girl from Luweero, weight 7.2 kg (WAZ −2.8, MUAC 11.9 cm), partially immunised (no MR), sleeps under a net 'sometimes', brought on day 4 of fever and cough after treatment at a drug shop, now with chest indrawing, SpO_2 88% and a positive malaria RDT."

IMCI classification (worked example):

- Cough + chest indrawing (no danger sign) + SpO_2 88% → **severe pneumonia** (WHO 2024: hypoxaemia counts as severe) → pink: oxygen, injectable antibiotics, admit/refer.
- Fever + RDT positive in a high-malaria area → **malaria**; with severe pneumonia, treat as **severe febrile illness** with artesunate if any danger sign, otherwise ACT.

- MUAC 11.9 cm, no oedema → **moderate acute malnutrition** → feeding counselling, supplementary feeding, follow up.
- Missed MR → **vaccinate before discharge**.
- Three delays: **first** (drug shop for 3 days), **second** (distance?), **third** (no oxygen at the first facility?).

Model case presentation (child):

> "Nakato is a 10-month-old girl from Kayunga, weight 6.4 kg (WAZ −2.7), length 69 cm (LAZ −1.5), WLZ −2.6, MUAC 12.0 cm, no oedema. She was born at home, is HIV-unexposed and is still breastfed, but complementary feeds are thin porridge twice a day. Her card shows vaccines to 14 weeks only; she has never had vitamin A or the malaria vaccine. She presented on day 5 of fever and day 2 of diarrhoea, after two visits to a drug shop where she was given 'syrups'. On arrival she had some dehydration and a positive malaria RDT. Summary: uncomplicated malaria with diarrhoea and some dehydration in a child with moderate wasting and missed immunisations, illustrating a first delay and several missed preventive opportunities. Plan: ACT 1 tablet twice daily for 3 days, ORS Plan B 480 mL over 4 hours, zinc 20 mg daily for 10 days, nutrition support, catch-up vaccines and vitamin A before discharge, ITN, and counselling on danger signs."

5. Clinical Acumen

Thinking like a clinician

- **Classify, don't diagnose only.** IMCI asks "How sick?" before "What disease?" A pink classification means **pre-referral treatment and referral now**, whatever the diagnosis.
- **One illness rarely comes alone.** Malaria + pneumonia + malnutrition is common. Assess every child for **all** main symptoms, nutrition, immunisation and HIV.
- **Hypoxaemia is invisible without a pulse oximeter.** Cyanosis is a late sign. Measure SpO_2 in every child with cough or difficult breathing.
- **Most hospital deaths happen in the first 24 hours.** Triage on arrival (ETAT), not after the queue.
- **"Treat and send home" includes counselling.** The mother must know **when to return immediately** (cannot drink, becomes sicker, fever, fast or difficult breathing, blood in stool) and when to come for **follow-up** (2 days for pneumonia/malaria, 5 days for persistent diarrhoea, 14 days for nutrition).
- **Use the three delays on every death and every near-miss**, then act at each level (MPDSR).

Red flags

RED FLAG

- **ETAT emergency signs**: obstructed breathing, central cyanosis, severe respiratory distress; shock (cold hands + CRT >3 s + weak fast pulse); coma (AVPU P or U); convulsions; severe dehydration with lethargy, sunken eyes and very slow skin pinch (in diarrhoea).
- **IMCI general danger signs**: unable to drink/breastfeed, vomits everything, convulsions, lethargic/unconscious.
- **Young infant (0–2 months) danger signs**: not feeding well, convulsions, RR ≥60, severe chest indrawing, temperature ≥37.5°C or <35.5°C, movement only when stimulated or none.
- **Severe acute malnutrition with medical complications.**
- **Hb <5 g/dL**, or <7 g/dL with respiratory distress.
- **A child returning** to the facility for the same illness.

The syndromic "differential": which killer is behind the presentation?

In child survival the "differential" is the **cause of death or severe illness** behind a presenting syndrome, and they often overlap.

Presenting syndrome	Main causes to distinguish	Distinguishing features	Key test
Fever without localising signs	Malaria, sepsis, UTI, typhoid, meningitis, HIV-related infection	RDT result, neck stiffness, urine, toxicity	RDT, urine dipstick, LP, blood culture
Fast breathing	Pneumonia, malaria with acidosis, severe anaemia, heart failure, sepsis	Crackles/indrawing vs deep (acidotic) breathing vs pallor, hepatomegaly	SpO_2, RDT, Hb, CXR
Diarrhoea	Acute watery (rotavirus, cholera), dysentery (Shigella), persistent (malnutrition, HIV, Giardia)	Duration, blood, dehydration, nutrition	Stool microscopy/culture
Convulsions with fever	Febrile seizure, cerebral malaria, meningitis, hypoglycaemia	Age, consciousness after the fit, RDT, glucose	RBS, RDT, LP
Newborn not feeding	Sepsis, hypothermia, hypoglycaemia, asphyxia/HIE, jaundice, prematurity	Temperature, glucose, birth history	RBS, temperature, bilirubin
Severe pallor	Malaria, sickle cell disease, iron deficiency, hookworm, HIV, malignancy	Spleen, jaundice, ethnicity/family history	Hb, smear, sickling test

Common pitfalls

- Treating a negative RDT as malaria, or treating every diarrhoea with antibiotics.
- Giving large fluid boluses to febrile children without true shock (FEAST).
- Discharging without catching up vaccines, vitamin A and deworming.
- Forgetting the newborn: a quarter of the ward's deaths may be in the first week of life.

6. Common Problems and Their Management

A. High-impact interventions along the continuum of care

Stage	Interventions (Uganda)
Adolescence and pre-pregnancy	Girls' education, family planning (birth spacing ≥24 months), HPV vaccine, nutrition, prevention of adolescent pregnancy, PMTCT readiness
Pregnancy	**ANC (WHO: 8 contacts)**, IPTp-SP, ITN, iron–folic acid, Td, syphilis and HIV testing (eMTCT), pre-eclampsia screening, birth preparedness
Birth	**Skilled attendance** at a facility, partograph/LCG, **Helping Babies Breathe** (Golden Minute), clean birth, **delayed cord clamping** (≥1 min), **antenatal corticosteroids** for preterm labour (24–34 weeks), management of complications
Newborn (first week)	**Immediate drying and skin-to-skin**, **breastfeeding within 1 hour**, **vitamin K**, tetracycline **eye ointment**, **chlorhexidine 7.1% cord gel**, BCG/OPV0/HepB-BD, **KMC** (immediate KMC for all babies <2 kg, WHO 2022), postnatal checks (within 24 h, day 3, days 7–14, 6 weeks), management of sepsis, jaundice and hypothermia

Stage	Interventions (Uganda)
Infancy and childhood	**Exclusive breastfeeding to 6 months**, complementary feeding from 6 months, **immunisation** (incl. PCV, rota, malaria vaccine), **vitamin A** and **deworming**, **ITNs**, growth monitoring and **CMAM** for wasting, **ORS + zinc**, **amoxicillin** for pneumonia, **RDT + ACT** for malaria, **rectal artesunate** pre-referral, **oxygen**, HIV testing and treatment, sickle cell screening, early childhood development, injury prevention
Environment	**Safe water, sanitation and handwashing (WASH)**, clean cooking fuel (less indoor smoke), housing

B. Delivering the interventions: platforms

1. **Community (VHTs/CHEWs, iCCM)**: health promotion, newborn home visits, case management of malaria, pneumonia and diarrhoea in children 2–59 months, referral, follow-up of immunisation defaulters.
2. **Health centres (IMCI/IMNCI)**: assess–classify–treat–counsel–follow up; immunisation; growth monitoring; ANC/PNC.
3. **Hospitals (ETAT+)**: triage, emergency care (oxygen, fluids, glucose, anticonvulsants), inpatient management of severe illness, newborn units (KMC, CPAP), blood.
4. **Campaigns**: measles–rubella, polio, vitamin A and deworming in **Child Health Days** (April and October in Uganda).
5. **Accountability**: **MPDSR** (every maternal and perinatal death reviewed), HMIS/DHIS2 data, verbal autopsy.

C. Key treatments and doses (first-level care)

Condition	Drug	Dose	Route	Frequency	Duration	Max / notes
Pneumonia (fast breathing / chest indrawing, no danger sign)	Amoxicillin (dispersible 250 mg)	**40 mg/kg/dose** (WHO 2024; = 80 mg/kg/day)	Oral	Twice daily	**3 days** fast-breathing pneumonia (WHO; low HIV settings), **5 days** chest indrawing; Uganda iCCM: **5 days** for all	iCCM: 2–11 months 1 tab, 1–5 years 2 tabs; max 1 g per dose
Severe pneumonia	Ampicillin (or benzylpenicillin) **+ gentamicin**	Ampicillin **50 mg/kg**; gentamicin **7.5 mg/kg**	IV/IM	Ampicillin 6-hourly; gentamicin once daily	At least 5 days	Ceftriaxone 80–100 mg/kg/day if failure; oxygen if SpO_2 <90%
Diarrhoea, no dehydration (Plan A)	Low-osmolarity **ORS**	<2 years 50–100 mL, ≥2 years 100–200 mL after each loose stool	Oral	After each stool	Until diarrhoea stops	Continue breastfeeding and feeding
Some dehydration (Plan B)	ORS	**75 mL/kg**	Oral	Over **4 hours**	Then reassess	

Condition	Drug	Dose	Route	Frequency	Duration	Max / notes
Severe dehydration (Plan C)	Ringer's lactate (or normal saline)	**100 mL/kg**: infants 30 mL/kg over 1 h then 70 mL/kg over 5 h; ≥12 months 30 mL/kg over 30 min then 70 mL/kg over 2.5 h	IV	–	Reassess every 1–2 h	**Not** this regimen in SAM (use ReSoMal, slower)
Diarrhoea (all)	**Zinc** (20 mg tablets)	2–6 months **10 mg (½ tab)**; 6 months–5 years **20 mg (1 tab)**	Oral	Once daily	**10 days** (UCG 2023; WHO 10–14 days)	WHO 2024 suggests 5 mg daily (section 7)
Dysentery	Ciprofloxacin	**15 mg/kg**	Oral	Twice daily	3 days	Max 500 mg per dose
Uncomplicated malaria	**Artemether–lumefantrine** 20/120 mg	5–14 kg **1 tab**; 15–24 kg **2**; 25–34 kg **3**; ≥35 kg **4**	Oral with fatty food/milk	At 0, 8 h, then twice daily	**3 days** (6 doses)	Infants <5 kg: treat per UCG
Severe malaria	**Artesunate**	**3 mg/kg** if <20 kg; **2.4 mg/kg** if ≥20 kg	IV (or IM)	At 0, 12, 24 h, then daily	Until able to take oral, then full ACT course	
Pre-referral (severe malaria, community/HCII)	**Rectal artesunate**	About **10 mg/kg**; UCG 2023 iCCM table by age: 4–11 months **1 capsule**, 1–3 years **2**, 3–5 years **3** (check the capsule strength in your pack, 50 or 100 mg); the UCG 2023 malaria chapter gives **100 mg** (5–<14 kg, 4 months–<3 years) and **200 mg** (14–19 kg, 3–<6 years), as in *Malaria in Children*	Rectal	Once	Then refer immediately	If expelled within 30 min, repeat
Vitamin A (prevention)	Retinol	6–11 months **100,000 IU**; 12–59 months **200,000 IU**	Oral	Every 6 months	–	Measles/SAM: treatment schedule
Deworming	Albendazole	12–23 months **200 mg**; ≥2 years **400 mg**	Oral	Single dose	Every 6 months	Or mebendazole 500 mg single dose (≥1 year)
Newborn	Vitamin K (phytomenadione)	**1 mg** (0.5 mg if <1.5 kg)	IM	Once	At birth	
Newborn cord	Chlorhexidine digluconate 7.1% gel	Apply to stump	Topical	Once daily	7 days (or once at birth per local protocol)	

Condition	Drug	Dose	Route	Frequency	Duration	Max / notes
Possible serious bacterial infection (young infant), referral refused or impossible (WHO 2015 PSBI)	Gentamicin + oral amoxicillin	Gentamicin **5–7.5 mg/kg**; amoxicillin **50 mg/kg/dose**	IM; oral	Gentamicin once daily; amoxicillin twice daily	Clinical severe infection: gentamicin **2 days** (or 7) + amoxicillin **7 days**; fast breathing alone (7–59 days): amoxicillin alone for 7 days	Only when referral is not possible; review on day 4
Hypoglycaemia	10% dextrose	**5 mL/kg**	IV (or sugar water by NGT)	Once, recheck	–	Threshold <2.5 mmol/L (<3 mmol/L in SAM)

DRUG DOSING

Doses here are summaries for first-level care. For full inpatient management see the topic pages (*Pneumonia*, *Diarrhoea & Dehydration*, *Malaria*, *Malnutrition*, *Neonatal Sepsis*). Always weigh the child.

UGANDA CONTEXT

Uganda's iCCM uses **colour-coded packs**: amoxicillin **red** (2–11 months) and **green** (1–5 years), ACT **yellow** (4 months–2 years) and **blue** (2–5 years). VHTs use a **respiratory timer** to count breaths and refer any child with a **danger sign**, chest indrawing or age under 2 months.

D. Emergency algorithm: the sick child arriving at a facility (ETAT)

1. **Triage** within minutes of arrival: emergency, priority or queue.
2. **Airway/Breathing**: position, suction, **oxygen** if SpO_2 <90% or central cyanosis, bag-mask ventilation if not breathing.
3. **Circulation**: shock (cold hands + CRT >3 s + weak fast pulse): **10–20 mL/kg Ringer's/saline** over 30–60 min (WHO; avoid large boluses in severe malaria/febrile illness without shock); in **SAM** 15 mL/kg over 1 h of a dextrose-containing solution; transfuse if Hb <5 g/dL.
4. **Coma/Convulsions**: recovery position, **glucose** (10% dextrose 5 mL/kg), **diazepam 0.5 mg/kg rectal** or 0.3 mg/kg IV (or midazolam 0.3–0.5 mg/kg buccal, max 10 mg), phenobarbital 20 mg/kg if fits continue.
5. **Dehydration (severe)**: Plan C.
6. **Then**: full assessment, specific treatment, monitoring chart, nutrition, immunisation check.

E. Prevention at every encounter ("no missed opportunity")

1. Check and update **immunisation**.
2. **Vitamin A** and **deworming** if due.
3. **Growth** (weight, height, MUAC) and feeding counselling.
4. **HIV** status known?
5. **ITN** at home? (distribution through ANC, EPI, mass campaigns).
6. **Counsel** on danger signs and when to return.

F. Consequences of failed child survival

At population level, the "complications" of failed child survival are:

- **Deaths** (the indicator), and **survivors with disability**: cerebral palsy after asphyxia or meningitis, epilepsy and cognitive impairment after cerebral malaria, blindness after measles/vitamin A deficiency, deafness after meningitis.
- **Stunting** with lifelong lower cognitive and economic potential.
- **Chronic lung disease** after severe pneumonia, measles, TB.
- **Family impact**: poverty from care costs, more pregnancies to replace lost children (high fertility).
- **Antimicrobial resistance** from untested fevers treated with antibiotics at drug shops.

7. Guidelines, Policy and Law in Uganda

Guideline or policy	What it sets
Uganda Clinical Guidelines 2023 (childhood illness, IMNCI, iCCM)	National case management, doses and referral rules for first-level care
WHO guideline on pneumonia and diarrhoea in children up to 10 years (2024)	Current WHO standards: hypoxaemia counts as severe pneumonia; amoxicillin 40 mg/kg/dose twice daily; zinc dose
WHO Pocket Book of Hospital Care for Children (2013) and **ETAT**	Triage and emergency treatment in hospitals
WHO PSBI guideline (2015)	Outpatient antibiotics for young infants when referral is not feasible
WHO recommendations for care of the preterm or low-birth-weight infant (2022)	Immediate KMC for all babies <2 kg
WHO ANC model	8 antenatal contacts
UNEPI routine schedule	Includes PCV, rotavirus and, since 2025, the **malaria vaccine** (MoH flag-off to 105 districts) Verify *current malaria vaccine schedule and district coverage*
RMNCAH Sharpened Plan	Uganda's investment plan for reproductive, maternal, newborn, child and adolescent health
MPDSR	Every maternal and perinatal death notified, reviewed and responded to
Child Health Days	Vitamin A, deworming and catch-up immunisation in April and October
SDG 3.2	NMR ≤12 and U5MR ≤25 per 1,000 live births by 2030

UGANDA CONTEXT

Where UCG and WHO differ.

- **Amoxicillin duration**: WHO 2024 gives **3 days** for fast-breathing pneumonia (low HIV settings) and **5 days** for chest indrawing; Uganda iCCM uses **5 days** for all.
- **Zinc**: UCG 2023 gives **10 mg** (2–6 months) and **20 mg** (6 months–5 years) for 10 days; WHO 2024 suggests **5 mg daily**. Follow UCG and the tablets you stock.
- **Rectal artesunate**: the UCG 2023 iCCM age table (1, 2 or 3 capsules) and the malaria chapter (100 mg or 200 mg by weight) express the dose differently; check the capsule strength in your pack.
- **Fluid boluses**: follow WHO/ETAT and the FEAST trial: boluses only for true shock, and cautiously.

8. Key Points

High-yield points

- U5MR is a **probability per 1,000 live births**; Uganda about **52** (UDHS 2022), NMR **22**.
- **About 40%** of under-five deaths in Uganda are **neonatal**, and this share is rising.
- Malaria, pneumonia and diarrhoea kill most children aged 1–59 months; **malnutrition** underlies about **45%**.
- Children die from **delays**: deciding, reaching, receiving care.
- IMCI asks **"How sick?"** before **"What disease?"**: pink = pre-referral treatment and refer.
- Measure **SpO_2** in every child with cough or difficult breathing.
- **Test before treating malaria**; antibiotics for diarrhoea only for dysentery or cholera.
- **No missed opportunity**: vaccines, vitamin A, deworming, growth and HIV status at every visit.
- First-hour newborn care: dry, skin-to-skin, breastfeed, vitamin K, eye care, cord care, KMC if <2 kg.
- Boluses only for true shock (FEAST).

Numbers & doses

Item	Value
Uganda NMR / IMR / U5MR	About **22 / 36 / 52** per 1,000 live births (UDHS 2022)
Uganda U5MR in 2016	**64** (UDHS 2016); about 90 in 2011
SDG 3.2 targets (2030)	NMR **≤12**; U5MR **≤25**
Global under-five deaths (2024)	**4.9 million**; **47%** neonatal
Global / sub-Saharan U5MR (2024)	**37.4 / 71.6** per 1,000
Undernutrition as underlying cause	About **45%** of child deaths
Lancet 2003	**23 interventions** could prevent **63%** of deaths
Stunting, Uganda	About **24%** (UDHS 2022)
Fast breathing	**≥60** (<2 m), **≥50** (2–11 m), **≥40** (1–5 y)
Hypoxaemia	SpO_2 **<90%**
Amoxicillin (pneumonia)	**40 mg/kg/dose twice daily**; iCCM 250 mg: 1 tab (2–11 m) or 2 tabs (1–5 y) 12-hourly for 5 days
Zinc	**10 mg** (2–6 months) / **20 mg** (6 months–5 years) daily for **10 days** (UCG 2023); WHO 2024 suggests **5 mg daily**
ORS Plan B	**75 mL/kg over 4 h**
Plan C	**100 mL/kg** Ringer's (30 then 70 mL/kg; over 6 h in infants, 3 h if ≥12 months)
AL bands	5–14 kg 1; 15–24 kg 2; 25–34 kg 3; ≥35 kg 4 tabs, twice daily × 3 days
IV artesunate	**3 mg/kg** (<20 kg); **2.4 mg/kg** (≥20 kg)
Rectal artesunate	About **10 mg/kg** once, then refer
Vitamin A	**100,000 IU** (6–11 m); **200,000 IU** (12–59 m) every 6 months
Albendazole	**200 mg** (12–23 m); **400 mg** (≥2 y) every 6 months
Vitamin K at birth	**1 mg** IM (0.5 mg if <1.5 kg)

Item	Value
Hypoglycaemia treatment	**10% dextrose 5 mL/kg**
Shock bolus	**10–20 mL/kg** over 30–60 min; SAM **15 mL/kg** over 1 h
MUAC severe / moderate	**<11.5 cm** / 11.5–12.4 cm
WHO ANC contacts	**8**
Postnatal checks	Within **24 h**, day **3**, days **7–14**, **6 weeks**
Follow-up after IMCI treatment	**2 days** (pneumonia, malaria), **5 days** (persistent diarrhoea), **14 days** (nutrition)

Exam traps

- **"Most under-five deaths in Uganda are from malaria."** Malaria is the biggest single killer after the neonatal period, but **neonatal deaths** together make up about **40%**; do not forget the newborn.
- **"U5MR is deaths per 1,000 children under 5."** Wrong: it is per **1,000 live births** (a probability of dying before age 5).
- **"IMCI is a treatment protocol for health workers."** Only one of its **three** components; the others are health systems and family/community practices.
- **"Treat every fever as malaria."** Wrong: **test first** (RDT); treat negatives for other causes.
- **"Give antibiotics for all diarrhoea."** Wrong: only for **dysentery** (and cholera/severe cases); otherwise ORS + zinc.
- **"Cyanosis means hypoxaemia; no cyanosis means OK."** Wrong: use the **pulse oximeter**; cyanosis is late.
- **"Big IV boluses for every febrile child with poor perfusion."** The FEAST trial (East Africa, including Uganda) showed boluses increased deaths in febrile children **without** shock; give boluses only for true shock, and cautiously.
- **"Plan C for the malnourished child with diarrhoea."** No: ReSoMal and slower rehydration unless shocked.

Questions seniors ask

Viva questions

1. *Define U5MR and give Uganda's current value.* Probability of dying before age 5 per 1,000 live births; about 52 (UDHS 2022).
2. *What are the commonest causes of under-five death in Uganda?* Neonatal causes (prematurity, asphyxia, sepsis), malaria, pneumonia, diarrhoea, with malnutrition as an underlying cause.
3. *What are the three components of IMCI?* Health-worker skills, health systems, family and community practices.
4. *What does iCCM treat?* Malaria, pneumonia and diarrhoea in children 2–59 months, plus newborn home visits and referral.
5. *Explain the three delays.* Deciding to seek care, reaching care, receiving adequate care.
6. *What is GOBI-FFF?* Growth monitoring, ORS, Breastfeeding, Immunisation; Female education, Family spacing, Food supplementation.
7. *Name five newborn interventions that save lives.* Skilled birth care with HBB, early breastfeeding, thermal care/KMC, chlorhexidine cord care, prompt treatment of sepsis (also vitamin K, steroids for preterm labour).

On the ward round

Why has neonatal mortality fallen more slowly than post-neonatal mortality? Post-neonatal deaths respond to vaccines, nets, ORS and antibiotics delivered at community level; newborn deaths need **quality care around birth** (skilled attendants, resuscitation, thermal care, sepsis treatment, preterm care), which is harder to deliver.

What single intervention would you scale up first in Uganda, and why? Defensible answers: **quality care at birth and in the first week** (largest share of deaths); **oxygen and pulse oximetry**; **malaria prevention and prompt treatment** (largest single cause after the newborn period). Justify with burden, effectiveness and cost.

What is iCCM and what are its limits? VHTs test and treat malaria, pneumonia and diarrhoea in children 2–59 months with RDTs, ACTs, amoxicillin, ORS and zinc, and refer danger signs. Limits: stock-outs, supervision, VHT motivation (often unpaid), young infants (<2 months) must be referred.

How do you use the three delays to analyse a child's death? Ask when the illness started and when care was sought (delay 1), how long it took to reach a facility and why (delay 2), and what happened at the facility (triage, drugs, oxygen, staff) (delay 3); then plan actions at each level (MPDSR).

How does malnutrition change your management of a child with diarrhoea? Use **ReSoMal** and slower rehydration (not Plan C boluses unless shocked), give antibiotics routinely in SAM, treat hypoglycaemia and hypothermia, and start F-75 early.

Name the newborn interventions you would ensure in the first hour of life. Dry and stimulate, HBB if not breathing, skin-to-skin, delayed cord clamping, breastfeeding within 1 hour, vitamin K, eye prophylaxis, chlorhexidine cord care, and weigh (KMC if <2 kg).

Mnemonics

- **"Three Ds of delay"**: **D**ecide, **D**istance, **D**elivery of care.
- **GOBI-FFF**: **G**rowth monitoring, **O**RS, **B**reastfeeding, **I**mmunisation; **F**emale education, **F**amily spacing, **F**ood supplementation.
- **ETAT emergency signs "ABCD"**: **A**irway/**B**reathing, **C**irculation, **C**oma/**C**onvulsions, **D**ehydration.
- **Priority signs "3 TPR-MOB"**: Tiny baby, Temperature, Trauma; Pallor, Poisoning, Pain; Restless/lethargic, Respiratory distress, Referral; Malnutrition, Oedema, Burns.
- **IMCI danger signs "DVCL"**: can't **D**rink, **V**omits everything, **C**onvulsions, **L**ethargic.
- **Fast breathing "60-50-40"**: <2 months ≥60; 2–11 months ≥50; 1–5 years ≥40.
- **Child survival "PPD-M"** (the big killers): **P**reterm/perinatal, **P**neumonia, **D**iarrhoea, **M**alaria (and **M**alnutrition behind them all).

OSCE tips

- IMCI stations test **counting RR for 60 seconds**, **classifying by colour**, and **counselling** (how to give ORS, when to return).
- Say the classification aloud ("This is **severe pneumonia**, a pink classification, so I will give the first dose of antibiotic and refer urgently").
- In counselling stations, cover **how to give ORS** (small frequent sips, spoon for infants, continue breastfeeding), the **zinc** course, the signs to **return immediately**, and the **follow-up** day.
- Always finish with prevention: check the card, vitamin A, deworming, ITN.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define the under-five mortality rate (U5MR)	The probability of dying before the **5th birthday**, expressed **per 1,000 live births**.
2	Define neonatal and infant mortality rates	**NMR**: deaths in the first **28 days** per 1,000 live births. **IMR**: deaths before **1 year** per 1,000 live births.
3	Uganda's NMR, IMR and U5MR (UDHS 2022)	About **22**, **36** and **52** per 1,000 live births (UDHS 2016: 27, 43, 64).
4	SDG 3.2 targets for 2030	Every country to reduce **NMR to ≤12** and **U5MR to ≤25** per 1,000 live births.
5	Global under-five deaths in 2024 (WHO)	**4.9 million**; **47%** were neonatal; global U5MR **37.4**; sub-Saharan Africa **71.6** per 1,000 and **58%** of all deaths.
6	Leading causes of under-five death globally	**Preterm birth complications**, **intrapartum events** (asphyxia/trauma), **pneumonia**, **diarrhoea**, **malaria**.
7	Leading causes of neonatal death	**Prematurity**, **intrapartum asphyxia**, **sepsis/pneumonia** (about 8 in 10 newborn deaths), plus congenital anomalies.
8	What proportion of child deaths has undernutrition as an underlying cause?	About **45%**.
9	The three components of IMCI	1. Improve **health-worker case-management skills**. 2. Strengthen **health systems**. 3. Improve **family and community practices**.
10	IMCI general danger signs (2 months–5 years)	**Unable to drink or breastfeed**, **vomits everything**, **convulsions**, **lethargic or unconscious**. Any one = urgent referral.
11	IMCI colour classification	**Pink**: urgent pre-referral treatment and referral. **Yellow**: specific treatment at the facility/home. **Green**: home care and advice.
12	Fast-breathing thresholds	**≥60/min** under 2 months; **≥50/min** at 2–11 months; **≥40/min** at 12–59 months. Count for a full minute.
13	What does iCCM treat in Uganda, and in which age group?	**Malaria** (RDT + ACT), **pneumonia** (amoxicillin DT), **diarrhoea** (ORS + zinc) in children **2–59 months**; pre-referral rectal artesunate; newborn home visits; referral of danger signs. Since **2010**.
14	iCCM amoxicillin dose (250 mg dispersible tablet)	2–11 months **1 tab**, 1–5 years **2 tabs**, every **12 hours for 5 days**.
15	Zinc for diarrhoea (UCG 2023)	2–6 months **10 mg (½ tab)**; 6 months–5 years **20 mg (1 tab)** once daily for **10 days**.
16	ORS Plan B	**75 mL/kg over 4 hours**, then reassess.
17	The three delays model	1. Delay in **deciding to seek care**. 2. Delay in **reaching** care. 3. Delay in **receiving adequate care** at the facility.
18	GOBI-FFF	**G**rowth monitoring, **O**ral rehydration, **B**reastfeeding, **I**mmunisation; **F**emale education, **F**amily spacing, **F**ood supplementation.
19	Lancet Child Survival Series (2003) key finding	**23** low-cost interventions at full coverage could prevent about **63%** of child deaths.
20	ETAT emergency signs	**A**irway/**B**reathing (obstructed, cyanosis, severe distress), **C**irculation (shock), **C**oma/**C**onvulsions, severe **D**ehydration.

#	Question	Answer
21	Shock in ETAT (all three needed)	**Cold hands** + **capillary refill >3 s** + **weak, fast pulse**.
22	What did the FEAST trial show?	In African children with severe febrile illness and impaired perfusion **without** hypotensive shock, **fluid boluses increased mortality**. Use boluses only for true shock.
23	Newborn interventions in the first hour	Dry and stimulate (HBB if not breathing), **skin-to-skin**, **delayed cord clamping**, **breastfeeding within 1 h**, vitamin K, eye ointment, chlorhexidine cord care, weigh (KMC if <2 kg).
24	Kangaroo mother care: current WHO recommendation (2022)	**Immediate KMC** for all babies **<2 kg** (continuous skin-to-skin started as soon as possible after birth, even before stabilisation).
25	WHO recommended number of ANC contacts	**8** contacts (WHO 2016).
26	Postnatal check schedule	Within **24 hours**, on day **3**, days **7–14**, and at **6 weeks**.
27	Vitamin A supplementation doses	6–11 months **100,000 IU**; 12–59 months **200,000 IU** every 6 months (Child Health Days in April and October).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy (10 kg) is brought to a HC III with 2 days of cough. He is drinking, has no vomiting or convulsions, and is alert. RR 46/min, no chest indrawing, no stridor, SpO_2 96%, temperature 37.9°C, malaria RDT negative. How should he be classified and managed under IMCI?

A Severe pneumonia: give the first dose of IM ampicillin and refer
B Cough or cold: home care with a soothing remedy
C Malaria: give artemether–lumefantrine
D Pneumonia: oral amoxicillin 40 mg/kg twice daily, counsel and review in 2 days
E Very severe febrile disease: give IM artesunate and refer

2. An 18-month-old girl (8.5 kg) died at home of pneumonia in a rural Ugandan district. During the death review, her mother says she noticed fast breathing on day 1 but waited 3 days because her husband was away with the money; when they reached the HC II, it had no amoxicillin, and they were referred but arrived at the hospital 8 hours later. Which delays contributed to this death?

A All three: deciding to seek care, reaching care and receiving care
B Only the first delay (deciding to seek care)
C Only the second delay (reaching care)
D Only the third delay (receiving adequate care)
E None: pneumonia deaths are not preventable in this setting

3. A 14-month-old boy (8.0 kg) has had watery diarrhoea for 2 days. He is restless and irritable, has sunken eyes, drinks eagerly, and his skin pinch goes back slowly. No blood in the stool. MUAC 13.4 cm, no oedema. What is the MOST appropriate management?

A Ringer's lactate 100 mL/kg IV over 3 hours (Plan C)
B Ciprofloxacin 15 mg/kg twice daily for 3 days
C ORS 75 mL/kg (600 mL) over 4 hours (Plan B), zinc 20 mg daily for 10 days, continue feeding
D ORS after each loose stool at home (Plan A) and return if worse
E Metronidazole and ORS

4. The DHO of a district with a U5MR of 70 per 1,000 asks you which statement about Uganda's child mortality is MOST accurate.

A Neonatal mortality has fallen faster than post-neonatal mortality since 2016
B Malaria vaccine alone will achieve the SDG target for U5MR
C U5MR is calculated per 1,000 children aged under 5 in the population
D Most under-five deaths in hospitals occur after the first week of admission
E Neonatal deaths now make up a growing share (about 40%) of under-five deaths

5. A 3-year-old girl (12 kg) is carried into Mengo's casualty. She is lethargic, has cold hands, a capillary refill of 4 seconds and a weak, fast pulse (170/min). She has had fever for 2 days; RDT positive. There is no history of diarrhoea, and she is not malnourished. What is the MOST appropriate immediate fluid management (WHO ETAT)? Advanced

A No fluids: FEAST showed that boluses kill children with malaria
B Ringer's lactate 10–20 mL/kg (120–240 mL) IV bolus, reassess, and treat severe malaria with IV artesunate
C Ringer's lactate 60 mL/kg over 15 minutes
D ORS 75 mL/kg over 4 hours by nasogastric tube
E 5% dextrose 100 mL/kg over 6 hours

6. A VHT in Kamuli sees a 7-month-old boy (7.2 kg) with fever for 1 day. He is feeding well and has no danger signs; RR 44/min, no chest indrawing. The malaria RDT is positive. What should the VHT do?

A Refer to the HC IV because all infants need facility care
B Give amoxicillin and ACT together
C Give artemether–lumefantrine from the yellow pack (1 tablet twice daily for 3 days) and follow up
D Give rectal artesunate and refer
E Give paracetamol only and repeat the RDT tomorrow

7. A 1.6 kg baby is born at 33 weeks at a HC IV in Mubende. She is breathing well, HR 150, temperature 36.2°C, and her mother is well. The nearest neonatal unit is 60 km away and there is no transport tonight. Which intervention will MOST reduce her risk of death?

A Place her in a cot under extra blankets away from the mother
B Give prophylactic IV ampicillin and gentamicin for 7 days
C Start formula feeds by bottle every 3 hours
D Refer her immediately by boda-boda
E Start continuous skin-to-skin Kangaroo Mother Care and early breastfeeding (with expressed breast milk by cup if needed)

8. Which statement about the Integrated Management of Childhood Illness (IMCI) strategy is correct?

A It has three components: health-worker skills, health-system strengthening, and family and community practices
B It is a hospital triage system for emergency signs
C It manages only malaria, pneumonia and diarrhoea
D It was designed for doctors in referral hospitals
E It classifies children by final laboratory diagnosis

9. A 3-week-old baby (3.1 kg) is brought to a HC III with poor feeding for 1 day, temperature 38.2°C and severe chest indrawing. Her mother refuses referral to the hospital ('no money, no one at home'); after counselling she still cannot go. What is the BEST management option? Advanced

A Send home with oral amoxicillin only and advise review in a week
B Give oral ciprofloxacin and paracetamol
C Give IM ceftriaxone once and discharge
D Give IM gentamicin 5–7.5 mg/kg once daily plus oral amoxicillin 50 mg/kg twice daily, with daily review (WHO PSBI regimen when referral is not possible)
E Give paracetamol and ask her to return if the fever persists

10. A district health team in Karamoja wants to reduce child deaths with a fixed budget. Data show high rates of wasting, malaria and measles. Which combination is MOST likely to reduce under-five mortality quickly?

A Building a new CT scanner at the regional referral hospital

B Community-based management of acute malnutrition (CMAM) with RUTF, measles–rubella catch-up, ITN distribution and iCCM

C Routine chest X-rays for all children with cough

D Training two paediatric surgeons

E A school deworming programme alone

11. A 9-month-old girl (6.8 kg) presents with cough and fast breathing. RR 58/min, mild chest indrawing, alert, feeding. SpO_2 is 87% in air. She is HIV-unexposed. What is the MOST appropriate classification and action?

A Pneumonia: oral amoxicillin for 5 days at home

B Cough or cold: home care

C Pneumonia: oral amoxicillin and review in 2 days, because chest indrawing alone is no longer 'severe'

D Bronchiolitis: salbutamol nebulisation and discharge

E Severe pneumonia because of hypoxaemia: oxygen, injectable ampicillin and gentamicin, admit

12. Which of these is a component of the 'FFF' in UNICEF's GOBI-FFF child survival strategy?

A Fluid resuscitation

B Family spacing

C Fever treatment

D Fortified flour

E Facility delivery

13. A 4-year-old boy (14 kg) has been unconscious for 2 hours after 3 days of fever in a village 40 km from the nearest hospital. A VHT reaches him; the malaria RDT is positive and he responds only to pain. What should the VHT do FIRST?

A Give pre-referral rectal artesunate and arrange urgent referral

B Give oral artemether–lumefantrine with milk

C Give oral amoxicillin and ORS

D Wait for him to wake up and then give ACT

E Give paracetamol and tepid sponging, then review tomorrow

14. In a ward audit at a regional referral hospital, 60% of child deaths occurred within 24 hours of admission, and many children waited over 2 hours in the outpatient queue. Which intervention is MOST likely to reduce these deaths? Advanced

A Adding a second daily ward round by the consultant

B Buying more broad-spectrum antibiotics such as meropenem

C Implementing ETAT triage at the hospital entrance with oxygen, glucose and fluids immediately available

D Extending outpatient clinic hours

E Building a new paediatric ward

15. A mother at a HC III in Mityana brings her 6-month-old son for review. He is well, exclusively breastfed, weight 7.4 kg (WAZ +0.1). Which set of preventive actions is MOST complete for today?

A Start cow's milk and stop breastfeeding to prepare for weaning

B Give the 9-month vaccines early to save a visit

C Deworm with albendazole 400 mg and give vitamin A 200,000 IU

D Counsel on starting complementary foods while continuing breastfeeding, give vitamin A 100,000 IU and malaria vaccine dose 1 (eligible district), check his card, and confirm he sleeps under an ITN

E Advise exclusive breastfeeding until 12 months

Answers & explanations

1. D · Pneumonia: oral amoxicillin 40 mg/kg twice daily, counsel and review in 2 days

D is correct: at 12–59 months, RR ≥40 is fast breathing; with no chest indrawing, danger signs or hypoxaemia, the classification is pneumonia (yellow): oral amoxicillin (40 mg/kg/dose twice daily, about 400 mg), home care advice and follow-up in 2 days. A is wrong: there is no danger sign, stridor or hypoxaemia, so it is not severe pneumonia. B is wrong: an RR of 46 at 2 years is fast breathing, so this is more than a cold. C is wrong: the RDT is negative, so antimalarials are not indicated. E is wrong: he has no general danger sign or stiff neck to suggest very severe febrile disease.

2. A · All three: deciding to seek care, reaching care and receiving care

A is correct: waiting 3 days for money is delay 1 (decision); the long journey after referral is delay 2 (reaching care); the stock-out at the HC II is delay 3 (receiving adequate care). B is wrong: it ignores the stock-out and the transport problem. C is wrong: it ignores the decision delay and the stock-out. D is wrong: it ignores the family and transport delays, which were just as important. E is wrong: pneumonia deaths are largely preventable with early amoxicillin and oxygen.

3. C · ORS 75 mL/kg (600 mL) over 4 hours (Plan B), zinc 20 mg daily for 10 days, continue feeding

C is correct: two signs of 'some dehydration' (restless/irritable, sunken eyes, drinks eagerly, slow skin pinch) without severe signs means Plan B: ORS 75 mL/kg over 4 hours, plus zinc 20 mg daily for 10 days (6 months–5 years) and continued feeding. A is wrong: Plan C is for severe dehydration (lethargy, unable to drink, very slow skin pinch). B is wrong: antibiotics are for dysentery (blood) or cholera, not acute watery diarrhoea. D is wrong: Plan A is for no dehydration. E is wrong: metronidazole has no role without a specific indication such as amoebiasis or giardiasis.

4. E · Neonatal deaths now make up a growing share (about 40%) of under-five deaths

E is correct: U5MR fell from about 64 to 52 between 2016 and 2022, but NMR fell only from about 27 to 22, so the newborn share keeps rising (about 40%). A is wrong: neonatal mortality has fallen more slowly. B is wrong: the malaria vaccine reduces but does not abolish malaria, and malaria is one of several major causes. C is wrong: U5MR is per 1,000 live births (a probability), not per 1,000 children under 5. D is wrong: most hospital deaths occur within 24–48 hours of admission, reflecting late arrival and triage delays.

5. B · Ringer's lactate 10–20 mL/kg (120–240 mL) IV bolus, reassess, and treat severe malaria with IV artesunate

B is correct: she has all three signs of shock (cold hands, CRT >3 s, weak fast pulse); WHO ETAT gives 10–20 mL/kg isotonic crystalloid, then reassesses, alongside IV artesunate 3 mg/kg for severe malaria and a glucose check. A is wrong: FEAST excluded hypotensive shock and showed harm from boluses in children with impaired perfusion but not shock; true shock still needs fluid. C is wrong: 60 mL/kg rapid boluses are not recommended in African febrile children (FEAST). D is wrong: ORS is for dehydration from diarrhoea and is too slow for shock. E is wrong: 5% dextrose is hypotonic and not a resuscitation fluid; 100 mL/kg is the Plan C dehydration regimen.

6. C · Give artemether–lumefantrine from the yellow pack (1 tablet twice daily for 3 days) and follow up

C is correct: iCCM allows VHTs to treat uncomplicated malaria in children 4 months–5 years after a positive RDT; the yellow pack (4 months–2 years) is 1 tablet every 12 hours for 3 days. A is wrong: only infants under 2 months or children with danger signs must be referred. B is wrong: RR 44 is not fast breathing at 7 months (threshold 50), so amoxicillin is not indicated. D is wrong: rectal artesunate is for danger signs (pre-referral), which he does not have. E is wrong: a positive RDT should be treated today; delay risks severe malaria.

7. E · Start continuous skin-to-skin Kangaroo Mother Care and early breastfeeding (with expressed breast milk by cup if needed)

E is correct: for babies under 2 kg, immediate KMC (WHO 2022) reduces mortality by preventing hypothermia, supporting breastfeeding and reducing infection; she is stable, so start it now. A is wrong: separating her from her mother and relying on blankets increases hypothermia risk. B is wrong: she has no signs of infection, and prophylactic antibiotics do not replace thermal care and feeding. C is wrong: breast milk is preferred and bottle-feeding is unsafe; use expressed breast milk by cup if she cannot suckle. D is wrong: an unsafe night journey on a motorbike risks hypothermia and harm; KMC can be started at the HC IV and transfer planned in the morning if needed.

8. A · It has three components: health-worker skills, health-system strengthening, and family and community practices

A is correct: WHO/UNICEF IMCI (1995) has these three components. B is wrong: that is ETAT. C is wrong: IMCI also covers fever/measles, ear problems, malnutrition, anaemia, HIV, immunisation and feeding; it is iCCM that focuses on three illnesses. D is wrong: it was designed for first-level health workers (HCs). E is wrong: it classifies by clinical signs (pink, yellow, green) to decide urgency, not by laboratory diagnosis.

9. D · Give IM gentamicin 5–7.5 mg/kg once daily plus oral amoxicillin 50 mg/kg twice daily, with daily review (WHO PSBI regimen when referral is not possible)

D is correct: this young infant has possible serious bacterial infection (fever, poor feeding, severe indrawing); when referral is impossible, WHO (2015) supports simplified treatment with IM gentamicin plus oral amoxicillin and close follow-up. A is wrong: oral amoxicillin alone is only for fast breathing in infants 7–59 days old, and weekly review is far too late. B is wrong: ciprofloxacin is not a recommended neonatal sepsis regimen. C is wrong: a single dose is inadequate for possible serious bacterial infection. E is wrong: fever with danger signs in a 3-week-old needs antibiotics now.

10. B · Community-based management of acute malnutrition (CMAM) with RUTF, measles–rubella catch-up, ITN distribution and iCCM

B is correct: the leading local causes (wasting, malaria, measles) each have proven, cost-effective community interventions: CMAM, measles–rubella vaccination, ITNs and iCCM. A is wrong: a CT scanner helps few children at high cost. C is wrong: routine CXR does not change outcomes for most pneumonia and wastes resources. D is wrong: surgical care matters but addresses a small share of child deaths. E is wrong: deworming alone has little effect on mortality.

11. E · Severe pneumonia because of hypoxaemia: oxygen, injectable ampicillin and gentamicin, admit

E is correct: SpO_2 <90% makes this severe pneumonia whatever the other signs; she needs oxygen and injectable antibiotics (ampicillin 50 mg/kg 6-hourly + gentamicin 7.5 mg/kg daily) as an inpatient. A is wrong: chest-indrawing pneumonia can be treated at home with amoxicillin only if there is no hypoxaemia or danger sign. B is wrong: she has fast breathing and indrawing. C is wrong: the hypoxaemia, not the indrawing, makes this severe. D is wrong: salbutamol does not treat hypoxaemic pneumonia, and a hypoxic infant must not be sent home.

12. B · Family spacing

B is correct: FFF stands for Female education, Family spacing and Food supplementation. A is wrong: fluid resuscitation is ETAT, not GOBI-FFF. C is wrong: fever treatment is not one of the Fs. D is wrong: food fortification is a nutrition intervention but the F is 'food supplementation'. E is wrong: facility delivery is an important newborn intervention but not part of the acronym.

13. A · Give pre-referral rectal artesunate and arrange urgent referral

A is correct: unconsciousness is a danger sign of severe malaria; iCCM gives rectal artesunate (about 10 mg/kg: 200 mg for 14–19 kg / 3 to <6 years per the UCG 2023 malaria chapter; count capsules by the strength in the iCCM kit) and refers immediately. B is wrong: an unconscious child cannot swallow safely and needs parenteral or rectal treatment. C is wrong: amoxicillin and ORS do not treat severe malaria. D is wrong: waiting risks death; cerebral malaria can kill within hours. E is wrong: fever control does not treat the cause, and delay is lethal.

14. C · Implementing ETAT triage at the hospital entrance with oxygen, glucose and fluids immediately available

C is correct: early deaths and long waits point to failure to identify and treat emergency signs on arrival; ETAT triage (implemented in many East African hospitals) reduces early inpatient mortality. A is wrong: a second consultant round does not help children who die before being seen. B is wrong: the problem is speed of assessment, not the spectrum of antibiotics; meropenem overuse drives resistance. D is wrong: longer clinic hours do not prioritise the sickest children. E is wrong: more beds do not fix triage delays.

15. D · Counsel on starting complementary foods while continuing breastfeeding, give vitamin A 100,000 IU and malaria vaccine dose 1 (eligible district), check his card, and confirm he sleeps under an ITN

D is correct: at 6 months, start complementary feeding with continued breastfeeding, give vitamin A 100,000 IU (6–11 months) and the first malaria vaccine dose in eligible districts, update immunisations and reinforce ITN use. A is wrong: breastfeeding should continue to 2 years and beyond; unmodified cow's milk is not a replacement at 6 months. B is wrong: MR and yellow fever are given at 9 months (early MR only in outbreaks, and it would not count). C is wrong: deworming starts at 1 year, and 200,000 IU is the dose for 12–59 months. E is wrong: breast milk alone is not enough after 6 months; delaying complementary food causes growth faltering.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Death review: a child who died of pneumonia

You attend the monthly perinatal and child death review at a general hospital in Luwero. The case: a 13-month-old girl, weight 7.0 kg (WAZ −2.7), died 3 hours after admission with severe pneumonia. She had been coughing for 6 days; the family first bought syrup at a drug shop, then visited a HC II on day 4 (no amoxicillin in stock; referred), and arrived at the hospital on day 6. At triage she waited 90 minutes; SpO_2 was not measured until 40 minutes after admission (78%), and the oxygen concentrator was shared with another child.

1. Analyse the death using the three delays.

> **Delay 1 (decision)**: 4 days at home/drug shop before formal care: low recognition of fast breathing as dangerous, drug shop first. **Delay 2 (reaching)**: 2 days from HC II referral to hospital (transport, money). **Delay 3 (receiving care)**: amoxicillin stock-out at HC II; 90-minute triage wait; late SpO_2 measurement; shared oxygen.

2. What underlying (contributing) factors do you notice?

> **Moderate-to-severe underweight** (WAZ −2.7) raising pneumonia mortality; check immunisation (PCV, Hib, MR) and HIV status; possible indoor smoke exposure; household poverty.

3. What should have happened at the HC II on day 4?

> IMCI assessment: if fast breathing only → oral amoxicillin (40 mg/kg/dose BD); if chest indrawing/danger signs → **first dose of antibiotic** (oral amoxicillin or IM ampicillin/gentamicin where available) **before referral**, with a referral note and transport arranged; a pulse oximeter if available. A stock-out should be reported and amoxicillin borrowed from a neighbouring facility.

4. What changes at the hospital would you recommend?

> **ETAT triage** at the door with a trained triage nurse; **pulse oximetry for every child with cough/difficult breathing**; enough **oxygen** (concentrators, cylinders, solar-powered oxygen) with flow splitters; emergency trolley with antibiotics ready; audit time-to-oxygen and time-to-antibiotic.

5. What community-level actions would prevent similar deaths?

> VHT **iCCM** strengthening (amoxicillin stock, respiratory timers), community education on danger signs and **fast breathing**, regulation and training of **drug shops**, PCV/Hib/MR coverage, nutrition programmes, and an emergency transport scheme (e.g. community savings or ambulance hotline).

6. How should the review be conducted so that it improves care?

> **No-blame**, confidential, focused on systems; agree specific actions with named people and deadlines; feed into **MPDSR** reporting to the district and MoH; follow up implementation at the next meeting.

Take-away: Child deaths are usually a chain of delays; reviewing each death with the three-delays lens, without blame, turns tragedies into concrete system fixes: triage, oxygen, stock and community recognition.

Case 2: IMCI at a HC III: the child with several problems

A 20-month-old boy is brought to a HC III in Mayuge with fever for 3 days and diarrhoea for 2 days. Weight 8.2 kg, length 79 cm, MUAC 12.1 cm, no oedema. He is drinking eagerly, not vomiting everything, no convulsions, alert. Temperature 38.6°C, RR 36/min, no chest indrawing. Eyes sunken, skin pinch goes back slowly. No blood in stool. Malaria RDT positive. His card shows vaccines up to 14 weeks only.

1. Check for general danger signs and classify each main symptom.

> **No general danger signs.** Cough/breathing: RR 36 at 20 months is normal (<40): **no pneumonia**. Diarrhoea: sunken eyes + slow skin pinch + drinks eagerly = **some dehydration**. Fever: RDT positive = **malaria**. Nutrition: MUAC 12.1 cm = **moderate acute malnutrition**. Immunisation: **incomplete** (missed MR1, YF, malaria vaccine, vitamin A).

2. What treatment will you give today, with doses?

> **ORS Plan B 75 mL/kg = about 600 mL over 4 hours** at the clinic; **zinc 20 mg** once daily for 10 days; **artemether–lumefantrine 1 tablet** (5–14 kg) at 0 and 8 h, then twice daily for 3 days; **paracetamol 15 mg/kg ≈ 120 mg** for fever; continue breastfeeding and feeding.

3. Which preventive actions belong to this visit?

> **Vitamin A 200,000 IU**, **albendazole 200 mg** (12–23 months) if not given in the last 6 months, **MR1 + yellow fever** today (minor illness is not a contraindication), malaria vaccine if the district is eligible, **HIV test** if status unknown, ITN check.

4. What will you tell the mother before she goes home?

> How to give ORS (small frequent sips; give more after each stool), complete the ACT with milk/fatty food, zinc for 10 days even after diarrhoea stops, feeding advice (energy-dense foods, extra meal daily), and **return immediately** if he cannot drink, becomes sicker, has blood in stool, convulses or is very sleepy.

5. When will you follow him up and for what?

> **2 days** for malaria (and diarrhoea if not improving), **14 days** for moderate acute malnutrition (weight gain, enrol in supplementary feeding if available), and next immunisation date (MR2 at 18 months already due: he is 20 months, so give MR2 4 weeks after MR1).

Take-away: IMCI assesses every child for all main symptoms, nutrition and immunisation, classifies each, treats them together and closes preventive gaps at the same visit.

Case 3: Planning newborn survival at a HC IV

You are an intern at a HC IV in Kiboga that conducts 250 deliveries a month. Last year there were 31 neonatal deaths: 12 from birth asphyxia, 10 in preterm/LBW babies (mostly hypothermia and feeding problems) and 7 from sepsis. The facility has one resuscitation bag-and-mask, no KMC area, and newborns are routinely bathed within an hour of birth.

1. Estimate the facility's neonatal mortality and comment.

> About **31 deaths / 3,000 births ≈ 10 per 1,000** facility-born babies. Lower than the national NMR (about 22), but these deaths are concentrated in **preventable** categories (asphyxia, hypothermia, sepsis), and community deaths are not counted.

2. How would you reduce asphyxia deaths?

> **Helping Babies Breathe** training for all midwives with monthly drills (the Golden Minute: dry, stimulate, clear airway if needed, ventilate within 1 minute), **one bag-and-mask per delivery bed** plus spares, functioning suction; **partograph/Labour Care Guide** with fetal heart monitoring to prevent intrapartum hypoxia; timely referral for CS.

3. How would you reduce deaths in preterm/LBW babies?

> Create a **KMC room**; start **immediate KMC** for every baby <2 kg (WHO 2022); **delay bathing for at least 24 hours**; early breastfeeding and expressed breast milk by cup; **antenatal corticosteroids** for women in preterm labour 24–34 weeks where criteria are met; warm room (25–28°C); temperature checks.

4. How would you reduce sepsis deaths?

> Clean birth practices and **handwashing**, **chlorhexidine 7.1% cord gel**, early exclusive breastfeeding, **recognise danger signs** (poor feeding, temperature, fast breathing, lethargy) and treat promptly with **ampicillin 50 mg/kg + gentamicin 5 mg/kg** (per neonatal protocol), and PSBI simplified regimens when referral is impossible.

5. How will you know whether the changes work?

> Track monthly **facility NMR**, **intrapartum stillbirths** (fresh stillbirths) and causes of death through **perinatal death reviews (MPDSR)**; process indicators: % babies <2 kg in KMC, % breastfed within 1 hour, % bathed after 24 h, HBB drill completion.

Take-away: Most newborn deaths at facility level are from asphyxia, hypothermia/prematurity and sepsis; HBB, immediate KMC, delayed bathing, early breastfeeding, clean cord care and death reviews tackle all three cheaply.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.